

TIM SANANIKONE

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EDUCATION

University of California, Irvine

Irvine, CA

B.S. in Computer Science; GPA: 3.74

Sept 2023 – June 2025

- **Relevant Coursework:** Project in Computer Vision, Neural Networks and Deep Learning, Machine Learning and Data-Mining, Artificial Intelligence, Data Structures and Algorithms, Embedded Software, Internet of Things (IoT) Software and Systems

SKILLS

Programming Languages: Python, C++, C, Assembly, MATLAB, Java, HTML/CSS

Computer Vision & Machine Learning: PyTorch, OpenCV, Segment Anything Model (SAM), YOLOv8, Neural Radiance Fields (NeRF), ML-based Segmentation, Data Augmentation

Embedded Systems: ESP32 Microcontroller Programming (C++), Hardware-in-the-Loop (HITL) Testing, Parallel Programming, Systems Design

EXPERIENCE

Machine Learning Intern

June 2025 – Aug 2025

Monterey Bay Aquarium Research Institute (MBARI)

Moss Landing, CA

- Developed deep learning pipeline integrating SAM and SAHI to generate kelp coverage calculations through segmentations from 42 MP UAV images, overcoming challenges of glint, lighting variation, and irregular structures [GitHub].
- Tested YOLOv8 as an initial approach, then pivoted to SAM after demonstrating better robustness for complex coastal imagery.
- Scaled pipeline to process 26,000+ aerial images across six visually distinct coastal sites.
- Developed a vegetation index-based annotation workflow to create a standardized ground truth dataset, enabling reproducible model evaluation and future benchmarking.

Undergraduate Researcher

Apr 2024 – June 2025

Molloy Lab, Department of Radiological Sciences

Irvine, CA

- Developed a 3D U-Net for airway segmentation of sub-2mm lung structures, addressing limited data with custom preprocessing and a patch-based training pipeline [GitHub].
- Implemented data augmentation (flips, deformations) to mitigate data scarcity and improve model generalization, achieving 94% dice accuracy with only 25 training volumes.
- Built digital phantom generation pipeline producing 50+ synthetic CT volumes, enabling robust testing of multi-material decomposition algorithms and increasing iodine map clarity by 20%.
- Created RAW-to-DICOM conversion tool standardizing lab workflows and improving compatibility across imaging studies.

PROJECTS

Light-Sensing Plant Robot — C++, ESP32, Sensor Processing

GitHub

- Built an autonomous light-seeking robot using an ESP32 microcontroller programmed in C++.
- Validated navigation algorithm using Hardware-in-the-Loop (HITL) testing by replaying real sensor data through a simulation to verify correct motor responses.

NeRF Model for 3D Reconstruction — Python, PyTorch, Volumetric Rendering

GitHub

- Implemented a Neural Radiance Fields (NeRF) model from scratch with hierarchical sampling and dual-network architecture.
- Achieved mean squared error of 0.0018, producing high-fidelity volumetric reconstructions from limited 2D input images.